

Exact Local Output Perturbation in KV-Cache Eviction: Ranking Inversions and Their Downstream Consequences

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Abstract

Deleting a token from a transformer’s KV cache and renormalizing the surviving attention weights changes a single attention head’s output by an amount that can be computed exactly, in closed form, before any downstream computation occurs. This raises a natural question: if the immediate local effect of an eviction decision is known exactly, how much does that exact local information tell us about the model’s eventual downstream behavior? We derive a closed-form, necessary-and-sufficient boundary condition (Γ) for when this exact local-perturbation ranking of eviction candidates disagrees with the simpler attention-mass ranking, and show that such disagreements are neither rare nor dominant, and that when they occur they almost always change the eviction decision and produce a measurable downstream effect, replicated across a GPT-NeoX/Pythia-160M and a GPT-2 (124M) grid. However, in a stratified, cluster-aware causal-tracing audit of 250 such cases, the token favored by the exact local metric at the moment of eviction is overturned by later transformer blocks in roughly a third of cases (32.4%, 95% CI [26.1%, 39.2%], cluster bootstrap), a rate not reliably distinguishable between the two architectures we tested. A sequence of falsification experiments — static directional value-vector geometry, propagation depth, logit-space representation, and state-dependent block sensitivity — fails to reduce this downstream behavior to a simple predictor. We conclude that an exact, provably correct local measure of eviction perturbation is not, by itself, a reliable guide to downstream model behavior, and we characterize precisely where and how that gap arises. The central identity underlying our local score is algebraically identical, in its single-token case, to an equation independently derived by prior KV-cache work [1]; our contribution is the pairwise ranking-inversion boundary built on top of it and the empirical/causal study of its downstream consequences, not the identity itself. All results are established on two small-to-moderate autoregressive language models and a deliberately stratified (not uniformly random) case sample; we make no claim about larger or instruction-tuned models, and no claim about which eviction policy one should deploy.

1 Introduction

Autoregressive transformer inference keeps a key-value (KV) cache of every previously seen token so that later positions can attend to earlier ones without recomputation. For long contexts this cache dominates memory use, and a large body of work proposes *eviction policies* that discard some cached tokens to bound memory — most commonly by ranking tokens according to a cheap proxy of importance, such as accumulated attention mass [2, 3, 4].

Attention mass alone does not fully specify what deleting a token actually does to the head’s output: a token can carry substantial mass yet sit close to the aggregated output (so removing it

barely moves anything), or carry little mass yet sit far from it (so removing it, after renormalization, moves the output by more than its mass alone would suggest). A different, more exact question is available but has received comparatively little attention: for a single attention head, deleting a token from the softmax and renormalizing the remaining weights changes that head’s output by an amount that can be written down *exactly*, in closed form, as a function of the deleted token’s attention weight *and* its value vector’s displacement from the head’s current output. This exact local quantity is not a heuristic and not an approximation; it is an algebraic identity. Because it depends on both quantities and attention-mass ranking depends on only one, the two rankings need not agree; the boundary condition Γ we derive (Section 4) makes precise exactly when they disagree, rather than leaving it as an empirical observation.

This paper asks a single, narrow question:

If the immediate effect of deleting a KV token can be computed exactly, how much does that exact local information tell us about downstream model behavior?

The answer we find, and defend with pre-registered, cluster-aware, cross-architecture evidence, is:

It determines the local ranking exactly and yields an exact boundary for disagreement with attention-mass ranking, but it does not reliably determine the downstream winner after subsequent transformer blocks.

What this paper is not. This is not a new KV-cache eviction algorithm, not a claim that any policy studied here is optimal, not a benchmark comparison against deployed systems such as H₂O or SnapKV, not a universal theorem about transformers in general, not an AI-safety contribution, and not a claim that downstream transformer behavior is chaotic or unpredictable. It is a precise, falsification-oriented mechanistic study of the gap between an exact local perturbation measure and downstream outcome.

Prior-art correction stated up front. The single-token special case of the exact identity we build on is algebraically identical to an equation independently derived in concurrent KV-cache work, KeepKV [1] (Section 2, Section 3). We state this explicitly here, rather than leaving it for related work, because it determines what in this paper can be claimed as new: not the identity, but the pairwise ranking-inversion boundary derived from it (Section 4) and the empirical/causal study of its downstream consequences (Sections 6 through 8).

Contributions.

1. A closed-form, necessary-and-sufficient pairwise boundary condition (Γ) determining when an attention-mass ranking and an exact local-output-perturbation ranking of two eviction candidates disagree (Theorem 2). To our knowledge, no prior work derives this condition.
2. A four-step, numerically verified empirical chain relating ranking inversions to measurable downstream consequences, replicated across two architectures (Section 6).
3. A cluster-aware causal-tracing audit ($n = 250$ stratified cases) showing that the immediate local-damage winner is overturned by downstream computation in roughly one third of cases, with uncertainty estimated by a cell-level cluster bootstrap rather than treating candidate pairs as independent (Section 7).

4. A four-stage falsification sequence (static geometry, propagation depth, functional/logit-space representation, state-dependent block sensitivity) that fails to reduce this downstream behavior to a simple predictor, including a positive causal finding (state-dependence exists) that is nonetheless *not* an explanation for the reversals (Section 8).

2 Related Work

Exact eviction perturbation. KeepKV [1] derives, as an intermediate step toward a bound on merging-based KV compression, the identity $o' = (o - \alpha_e v_e) / (1 - \alpha_e)$ for the head output after a single token e is deleted and the remaining weights renormalized. As we show in Section 3, this is algebraically identical to the singleton case of the identity we use, $\Delta_i = \alpha_i(o - v_i) / (1 - \alpha_i)$. We do not claim this identity as new. KeepKV uses it operationally, inside an EMA-based error bound for its merging procedure; it does not use it to define a per-token ranking score, and it does not derive a ranking-inversion condition against attention-mass ranking. Our block- B generalization (simultaneous multi-token eviction) is a direct linear extension of this same substitution and is not presented as a separate contribution.

Attention-mass and heuristic eviction scoring. H₂O [2] retains tokens by accumulated attention-mass “heavy hitters”; Scissorhands [3] exploits persistence of token importance across steps; StreamingLLM [4] identifies an attention-sink phenomenon at initial positions. We use attention-mass ranking, in the spirit of this line of work, as one of the two policies compared throughout this paper. We do not compare against these systems empirically on downstream task quality; our comparison is restricted to the two rankings themselves and their measured effect on next-token KL divergence.

Value-aware and bound-based scoring. CriticalKV [5] derives an upper *bound* on output perturbation using an L1 value-state norm, not an exact identity. A concurrent diagnostic framework [6] scores candidates by squared distance to a weighted-centroid approximation of the retained set’s output, not the true renormalized output. Neither derives an explicit ranking-inversion condition between two named scoring rules; both are structurally related to, but not equivalent to, the score we use (to our knowledge; see Section 3).

Causal tracing and logit-space projection. Our downstream audits (Sections 7–8) apply activation patching / causal-mediation analysis, in the sense established by causal tracing of factual associations in GPT-style models [7], and logit-lens-style projection of intermediate hidden states to vocabulary space, in the sense formalized by the Tuned Lens [8]. Both are established general techniques that we apply to a specific, narrow question (does an eviction policy’s local ranking survive propagation); we do not claim either technique itself as a contribution of this paper.

3 Exact Local Eviction Geometry

3.1 Setup

For a single attention head, let $\alpha \in \mathbb{R}_{>0}^n$ be the softmax attention-weight vector over n keys, $\sum_k \alpha_k = 1$, and let v_k denote the value vector already mapped into the quantity that is linearly combined to form the head’s output (i.e. $v_k = W_O^h v_k^{\text{raw}}$ in the usual notation, so that the output is

a plain weighted sum). The head output is

$$o = \sum_k \alpha_k v_k.$$

For a block B of tokens to be evicted, define the block mass, block contribution, and residual

$$m_B = \sum_{i \in B} \alpha_i, \quad o_B = \sum_{i \in B} \alpha_i v_i, \quad r_B = o_B - m_B o.$$

Theorem 1 (Exact delete-and-renormalize identity). *Deleting block B from the softmax and renormalizing the surviving weights to sum to 1 changes the head output by exactly*

$$\Delta_B = \frac{m_B o - o_B}{1 - m_B} = -\frac{r_B}{1 - m_B}, \quad m_B < 1.$$

Proof. The renormalized output after deleting B is $o' = \sum_{k \notin B} \frac{\alpha_k}{1 - m_B} v_k = \frac{1}{1 - m_B} \left(\sum_k \alpha_k v_k - \sum_{i \in B} \alpha_i v_i \right) = \frac{o - o_B}{1 - m_B}$. Then $\Delta_B = o' - o = \frac{o - o_B}{1 - m_B} - o = \frac{o - o_B - o(1 - m_B)}{1 - m_B} = \frac{m_B o - o_B}{1 - m_B}$, which equals $-r_B/(1 - m_B)$ by definition of r_B . *This is direct algebraic substitution, not an approximation.* $\square$

This identity, its proof, and the exclusion of the degenerate case $m_B \rightarrow 1$ (where the renormalized output is undefined, not merely ill-conditioned) are implemented and unit-tested in this project’s code (`src/kv_efficiency/local_algebra.py`), independent of model architecture.

3.2 Prior-art correction: the singleton case is not new

For a single evicted token i ($B = \{i\}$), $m_B = \alpha_i$ and $o_B = \alpha_i v_i$, so Theorem 1 specializes to

$$\Delta_i = \frac{\alpha_i(o - v_i)}{1 - \alpha_i}. \quad (*)$$

KeepKV [1] derives, for the output after deleting a single token e and renormalizing, $o' = (o - \alpha_e v_e)/(1 - \alpha_e)$. Taking $o' - o$:

$$o' - o = \frac{o - \alpha_e v_e}{1 - \alpha_e} - o = \frac{\alpha_e o - \alpha_e v_e}{1 - \alpha_e} = \frac{\alpha_e(o - v_e)}{1 - \alpha_e},$$

which is line-for-line identical to $(*)$, including sign. **We state plainly that the singleton exact-perturbation identity is not a novel contribution of this paper.** It is cited here, and used, but not claimed as new. Our block- B form (Theorem 1) is a direct linear generalization to simultaneous multi-token eviction; we present it as an extension, not a separate result.

3.3 Exact local output perturbation

Definition 1 (Exact local output perturbation). *For a single candidate token i , define*

$$d_i = \frac{\alpha_i}{1 - \alpha_i} \|o - v_i\| = \|\Delta_i\|.$$

d_i is the exact Euclidean norm of the change in *one attention head’s output*, computed before any downstream transformation, caused by deleting token i from the softmax and renormalizing the surviving weights. We use **exact local output perturbation** as the primary term for d_i throughout this paper. It is not global damage, not downstream importance, not causal importance in the sense of Section 2’s causal-tracing literature, and not an optimality claim of any kind; each of those would require evidence this paper explicitly does not claim to provide, and indeed the results of Sections 7–8 show that d_i does not reliably predict several of them.

4 Pairwise Ranking-Inversion Boundary

Attention-mass ranking orders candidates by α_i alone. Exact local output perturbation orders them by $d_i = [\alpha_i/(1 - \alpha_i)] g_i$, $g_i := \|o - v_i\|$. These two rankings need not agree, because d_i also depends on g_i , which attention-mass ranking ignores entirely. We derive the exact condition under which they disagree.

Theorem 2 (Pairwise ranking-inversion boundary). *Let i, j be candidates with $0 < \alpha_i < \alpha_j < 1$ and $g_i, g_j > 0$. Define*

$$G_{ij} = \frac{g_i}{g_j}, \quad M_{ij} = \frac{\alpha_j(1 - \alpha_i)}{\alpha_i(1 - \alpha_j)}, \quad \Gamma_{ij} = \log G_{ij} - \log M_{ij}.$$

Then

$$d_i > d_j \iff G_{ij} > M_{ij} \iff \Gamma_{ij} > 0,$$

and symmetrically with $=$ and $<$.

Proof. $d_i > d_j \iff \frac{\alpha_i}{1 - \alpha_i} g_i > \frac{\alpha_j}{1 - \alpha_j} g_j$. All four quantities $\alpha_i, \alpha_j, 1 - \alpha_i, 1 - \alpha_j$ are strictly positive by hypothesis ($0 < \alpha_i < \alpha_j < 1$), as are g_i, g_j , so the inequality is preserved on dividing both sides by $\frac{\alpha_j}{1 - \alpha_j} g_j > 0$:

$$d_i > d_j \iff \frac{g_i}{g_j} \cdot \frac{\alpha_i(1 - \alpha_j)}{\alpha_j(1 - \alpha_i)} > 1 \iff G_{ij} > M_{ij}.$$

Since both G_{ij} and M_{ij} are strictly positive, $\log$ is a strictly increasing bijection on $(0, \infty)$, so $G_{ij} > M_{ij} \iff \log G_{ij} > \log M_{ij} \iff \Gamma_{ij} > 0$. The equality and reverse cases follow identically with $=$ and $<$. $\square$

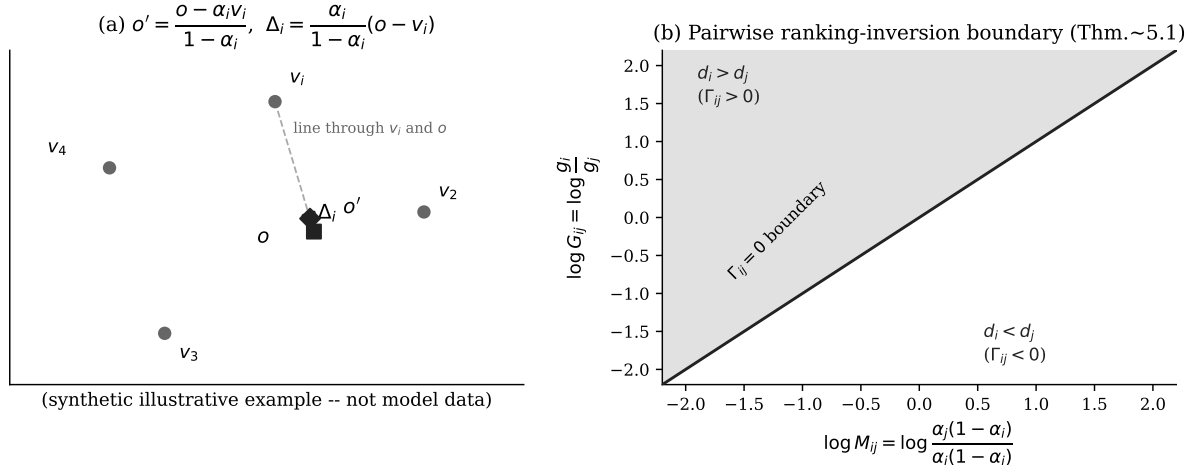
Figure 1 illustrates both the geometry of Theorem 1 and the boundary condition of Theorem 2.

Equality case. $\Gamma_{ij} = 0$ is the exact boundary at which the distance ratio G_{ij} exactly compensates the mass-ratio term M_{ij} ; on this boundary the two rankings agree in outcome even though they measure different quantities.

Edge cases. As $\alpha_i \rightarrow 0$, $d_i \rightarrow 0$ regardless of g_i : a vanishing-mass token does vanishing damage on removal, consistent with the identity, not an exception to it. The case $\alpha_i \rightarrow 1$ is excluded by hypothesis and by Theorem 1 (the renormalized output is undefined when a block carries essentially all the mass). When $\alpha_i = \alpha_j$, $M_{ij} = 1$ and $\Gamma_{ij} = \log(g_i/g_j)$: the mass term drops out and the comparison reduces to comparing g_i, g_j directly, as expected. When $g_i = 0$ or $g_j = 0$ (a candidate whose value vector coincides exactly with the head output), $d_i = 0$ or $d_j = 0$ trivially and $\log G_{ij}$ is undefined; the underlying (non-log) comparison $G_{ij} > M_{ij}$ remains well defined and should be used directly in this degenerate case.

Are logarithms necessary? No. The decisive statement is the ratio comparison $G_{ij} > M_{ij}$; Γ_{ij} is a monotone reparametrization chosen for numerical convenience (differences of logs avoid over/underflow when masses or distances are extreme, and give the boundary the natural form $\Gamma_{ij} = 0$). Γ_{ij} carries no mathematical content beyond G_{ij} and M_{ij} themselves.

No claim of novelty is made within this theorem statement or proof; the novelty assessment for this result is given in Section 2 and stated cautiously there (“to our knowledge”).



Note: panel (b) compares only the local-perturbation ranking of two candidates; it makes no claim about which eviction policy wins downstream (Sections 7-9).

Figure 1: (a) A schematic, synthetic example (not model data) of the exact delete-and-renormalize geometry of Theorem 1: deleting token i moves the head output from o to o' along the line through v_i and o , by exactly $\Delta_i = [\alpha_i / (1 - \alpha_i)](o - v_i)$. (b) The $\Gamma_{ij} = 0$ boundary of Theorem 2 in the $(\log M_{ij}, \log G_{ij})$ plane. Panel (b) compares only which of two candidates has the larger exact local perturbation; it makes no claim about which eviction policy produces the smaller downstream KL divergence, a separate, empirically measured question addressed in Sections 6–8.

5 Experimental Design and Validated Architecture Adapters

We evaluate all claims on two architectures: GPT-NeoX/Pythia-160M (parallel attention/MLP residual structure) and GPT-2 (124M) (sequential residual structure, Conv1D-based attention). To recompute exact per-head attention weights, value vectors, and outputs identically across both, we built two validated adapters exposing a shared contract (`src/kv_efficiency/attention_reconstruction*.py`). The GPT-2 adapter (new to this project) is validated by 8 dedicated unit tests plus inline numerical checks against a pilot grid (36 prompt $\times$ layer $\times$ head cells): row-sum deviation from 1 up to 3.28×10^{-7} , zero causal leakage, query/key recomputation error up to 3.27×10^{-7} , c_{proj} reconstruction relative error up to 1.97×10^{-7} , and delete-and-renormalize identity error up to 2.84×10^{-14} across 180 checks, all attributable to ordinary float32 accumulation, not an implementation defect. The full project test suite (74 tests, all passing at time of writing) includes these 8 GPT-2-adapter tests alongside the pre-existing architecture-independent local-algebra tests. This pilot grid is used exclusively for adapter validation in this paper; its own inversion-prevalence figures are not used as a headline empirical result anywhere in this manuscript (see Section 6 and Appendix B for the two grids that are).

Two grids were computed: a Pythia-160M grid (5 layers $\times$ 4 heads $\times$ 3 prompts, 55,971 total candidate pairs) and a GPT-2 full grid (12 layers $\times$ 12 heads $\times$ 3 prompts, 426,816 total candidate pairs). In both, every computed Γ_{ij} value was independently cross-checked against direct recomputation of d_i, d_j from the underlying α, v data: 0 mismatches in 55,971 checks (Pythia) and 0 mismatches in 426,816 checks (GPT-2). This is a sanity check on the implementation of Theorem 2, not evidence for any empirical claim.

Reproducibility. Pythia was loaded as EleutherAI/pythia-160m pinned to git revision 50f5173d932e

(12-character prefix of the full 40-character SHA recorded in the stored run provenance); GPT-2 was loaded as the unpinned Hugging Face `gpt2` checkpoint (no revision was pinned for GPT-2 in the underlying experiment code, unlike Pythia – noted here rather than obscured). Every stratified draw and bootstrap resample in this paper (the 250-case sample of Section 7, its cluster bootstrap, and the SDPH case selection of Section 8) uses the fixed seed 20,260,910. Every `results` directory cited in this paper has a corresponding, identically named script under `experiments/` in the project repository that produced it (for example, `results/downstream_propagation_sample/` is produced by `experiments/downstream_propagation_sample.py`). Exact prompt texts and grid configuration are recorded in `configs/milestone2_grid.yaml`.

6 Ranking Inversions and Their Immediate Consequences

We report four logically separate empirical statements, verified against the stored grid outputs and kept deliberately un-merged, since each has a different (and generally weaker, moving down the list) evidential strength.

- (1) **Attention-mass and exact-local-perturbation rankings sometimes invert.** Meaningful inversions ($\Gamma_{ij} \neq 0$ under the pre-declared floors) occur in $1,987/55,971 = 3.55\%$ of Pythia candidate pairs and $22,983/426,816 = 5.38\%$ of GPT-2 candidate pairs. The two grids are not equally dense or directly comparable (different layer/head coverage, different candidate-pool sizes per cell), so this prevalence difference should not be over-interpreted as an architecture effect; both figures show the same qualitative fact, that inversions are neither vanishingly rare nor dominant.
- (2) **Inversions almost always change the eviction selection.** Among these inversions, the evicted token set actually changes within the candidate range in 99.60% of Pythia cases and 99.57% of GPT-2 cases.
- (3) **Changed selections almost always produce a measurable downstream KL difference.** Among set-changing inversions, the resulting final-layer KL divergence exceeds the pre-declared, diagnostically justified noise floor in 99.14% of Pythia cases and 98.22% of GPT-2 cases. *Measurable* here means only “above the configured numerical threshold,” not necessarily practically large; advantage magnitudes span several orders of magnitude (Pythia interquartile range of |advantage| spans from $\sim 10^{-6}$ to $\sim 10^{-2}$; see stored percentile data).
- (4) **The locally preferred choice does not reliably remain the final downstream winner.** This is the subject of Section 7 and is *not* implied by (1)–(3).

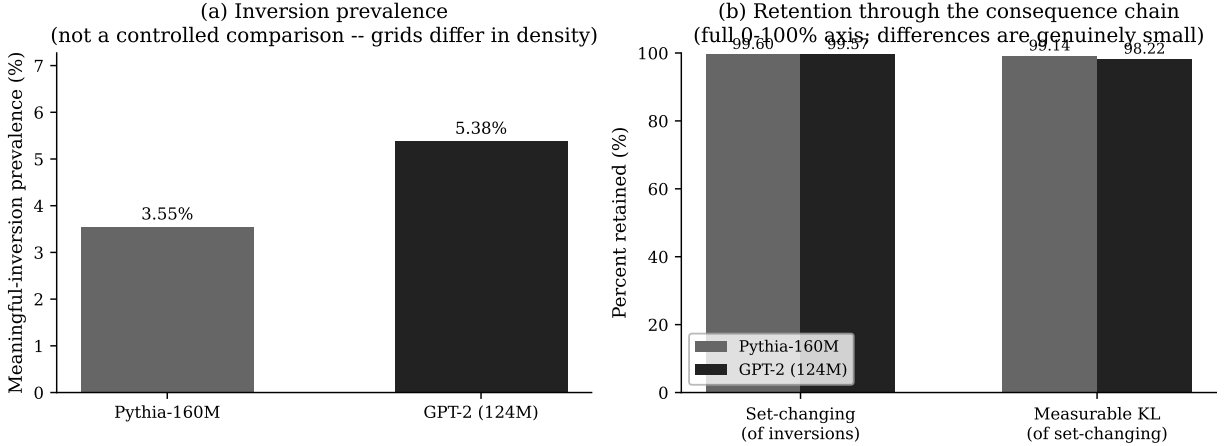
Figure 2 shows this chain numerically for both architectures.

Among measurable set-changing inversions, exact-local-perturbation ranking wins (produces the smaller final-layer KL) in 56.27% of Pythia cases ($n = 1,962$; Wilson 95% CI [54.1%, 58.5%]) and 57.34% of GPT-2 cases ($n = 22,477$; 95% CI [56.7%, 58.0%]). This is a modest majority, not a near-certainty, and is reported descriptively; it is not evidence that local-perturbation ranking is a better eviction policy in any downstream or task-quality sense, which this paper does not evaluate.

7 From Local Ranking to Downstream Outcome

7.1 Design

To test whether the immediate local winner (statement (4) above) survives propagation through later transformer blocks, we traced 250 candidate pairs (100 Pythia, 150 GPT-2) stratified by depth tertile and final winner, recording the sign of $d_{\text{mass}} - d_{\text{local}}$ in residual-stream distance at every downstream layer up to the output, and classifying each case as *immediate_consistent* (the sign at



Statements (1)-(3) of Section 7 only; downstream-winner reversal (statement 4) is Figure 3.

Figure 2: Statements (1)–(3) above, computed directly from the stored grid outputs (`results/full_grid_mechanistic_analysis/` for Pythia, `results/cross_architecture_full_grid/` for GPT-2; see `manuscript/figures/FIGURE_PROVENANCE.md` for the exact keys and filters). (a) Meaningful-inversion prevalence on its own scale; the two grids differ in density and coverage, so this is not a controlled comparison (Section 10). (b) Set-change and measurable-KL retention on the full 0–100% axis, so the genuinely small architecture differences are not visually exaggerated. Statement (4) — whether the locally preferred choice remains the downstream winner — is deliberately excluded from this figure; it is addressed in Figure 3 and Section 7.

the eviction layer matches the final winner throughout), *single_flip*, *multiple_flip*, *never_consistent*, or *unresolved*. This stratified sample governs *case selection*; it does not by itself establish *statistical independence* between cases, since multiple sampled cases can share the same (architecture, prompt, layer, head) cell and therefore the same underlying forward pass and attention distribution.

7.2 Cluster-aware uncertainty

Because of this shared structure, we treat the (architecture, prompt, layer, head) cell — not the individual candidate pair, and not the depth/winner sampling stratum used for case selection — as the clustering unit for uncertainty estimation, and obtain all reported intervals by a nonparametric cluster bootstrap (10,000 replicates, architecture-stratified: within each replicate, Pythia cells and GPT-2 cells are independently resampled with replacement, preserving each architecture’s observed cell count, before combining for pooled statistics). Reported pair counts ($n = 250, 100/150$ by architecture) describe computational coverage — how many candidate pairs were traced — while cluster-aware intervals describe uncertainty across repeated draws of structural context (attention cells); these are different quantities. The 250-case sample is stratified by design, so these rates should not be read as raw, unweighted population prevalence among all inversion pairs, only as estimates within the sampled design.

The clustering produced 155 pooled cells (34 Pythia, 121 GPT-2); 67.7% of pooled cells, 26.5% of Pythia cells, and 79.3% of GPT-2 cells contain exactly one sampled case (i.e. correction is modest for GPT-2’s finer-grained sample and larger, but still not extreme, for Pythia’s coarser one).

7.3 Results

	Pooled ($n = 250$)		Pythia ($n = 100$)		GPT-2 ($n = 150$)	
Flip rate	0.324	[0.261, 0.392]	0.380	[0.264, 0.522]	0.287	[0.216, 0.360]
Immediate-consistent rate	0.432	[0.367, 0.494]	0.380	[0.260, 0.485]	0.467	[0.388, 0.544]
Unresolved rate	0.244	[0.186, 0.307]	0.240	[0.136, 0.352]	0.247	[0.178, 0.320]

Table 1: Cluster-aware point estimates and 95% percentile bootstrap intervals ($B = 10,000$; cell = architecture $\times$ prompt $\times$ layer $\times$ head). Source: `results/cluster_aware_reanalysis/`.

The architecture difference in flip rate (Pythia – GPT-2) is 0.093, 95% CI $[-0.045, 0.252]$: this interval includes zero, so **no reliable architecture difference is established**; both architectures show substantial downstream reversal, and the point estimate favoring Pythia should not be promoted beyond that. Given the pooled estimate and its interval, describing downstream reversal as occurring in “**roughly one third**” of the stratified sample is defensible; describing it as rare, or as near-universal, is not.

Figure 3 shows one representative trajectory per category, illustrating that the local ordering at the eviction layer can be preserved, reversed once, reversed multiple times, or remain unresolved through to the final layer.

7.4 Local-vs-mass asymmetry

	Local-final-winner ($n = 126$)		Mass-final-winner ($n = 124$)	
Flip rate	0.206	[0.137, 0.279]	0.444	[0.347, 0.543]
Immediate-consistent rate	0.619	[0.519, 0.711]	0.242	[0.170, 0.320]

Table 2: Subgroup cluster-bootstrap estimates by final downstream winner (89 clusters for local-final-winner, 93 for mass-final-winner).

Cases where the exact local metric’s preferred choice is also the final downstream winner are, unsurprisingly, disproportionately *immediate_consistent*; cases where attention-mass ultimately wins are disproportionately associated with a flip. We report this as an **empirical asymmetry requiring explanation**, not as a mechanism; no experiment in this paper establishes why mass-final-winner cases flip more often, and Section 8’s SDPH result specifically rules out one tempting explanation (state-dependence) as the mechanism.

8 Falsification of Candidate Explanations

Given Section 7’s result, we ask what *does* explain, or at least correlate with, whether a given case’s immediate local winner survives to become the downstream winner. Four candidate explanations were tested, each motivated by the failure of the previous one; this is a directed, sequential falsification program, not four independent preregistered confirmatory trials, and we do not apply a multiple-comparisons correction across them (Section 9).

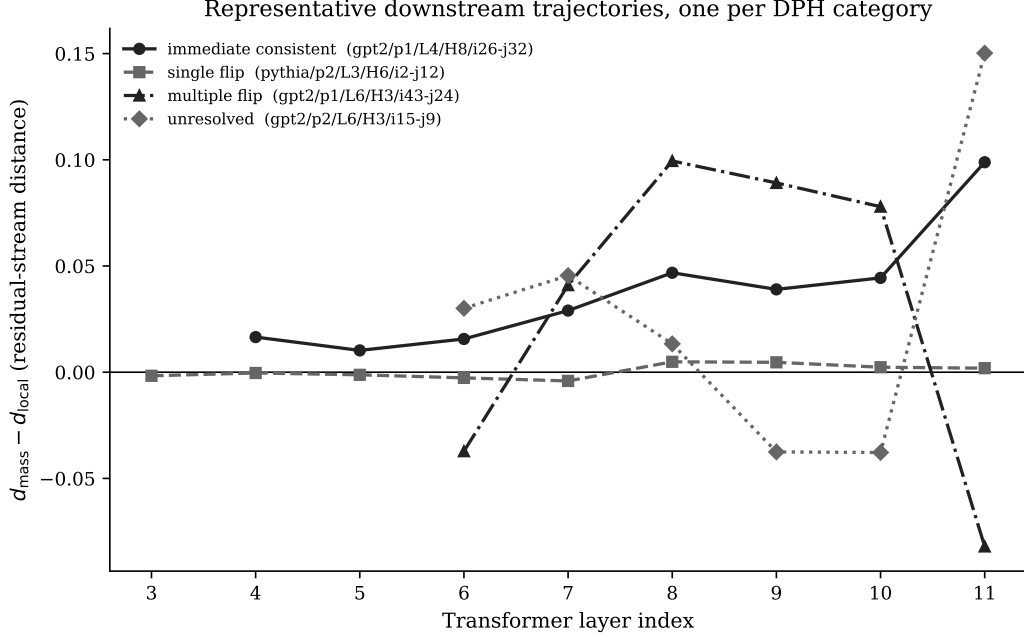


Figure 3: One representative case per DPH category, selected from the 250-case sample (`results/downstream_propagation_sample/cases.json`) by a fixed, documented rule: within each category, the case whose $|\text{advantage}_{\text{final}}|$ is closest to that category’s own median (i.e. a case of typical, not maximal, effect size; ties broken by a fixed architecture/prompt/layer/head/ i/j sort key). *never_consistent* and *unresolved* are pooled into one “unresolved” bucket, matching Table 1’s own convention. Selected cases: *immediate_consistent* = gpt2/p1/L4/H8/i26-j32; *single_flip* = pythia/p2/L3/H6/i2-j12; *multiple_flip* = gpt2/p1/L6/H3/i43-j24; *unresolved* = gpt2/p2/L6/H3/i15-j9 (full selection log, including each bucket’s size and median, in `manuscript/figures/FIGURE_PROVENANCE.md`). The plotted quantity is each case’s stored $d_{\text{mass}} - d_{\text{local}}$ residual-stream distance difference at every downstream layer, exactly as defined in Section 7; no aggregation across cases. These four trajectories are illustrative single examples ($n = 1$ per category), not distributional evidence; the distributional claims (flip rates and their uncertainty) are Table 1, not this figure.

8.1 DVG: static directional value-vector geometry

Eight cosine-similarity and norm-ratio features of the candidate value vectors (e.g. $\cos(v_i, v_j)$, $\cos(v_i, o)$, $\cos(o - v_i, o - v_j)$) were tested as predictors of the eventual winner, using pre-declared practical-weakness thresholds ($|\text{AUC} - 0.5| < 0.05$ and $|r| < 0.10$) rather than raw significance. On Pythia ($n = 1,962$ measurable pairs), the best grouped cross-validated model reaches AUC 0.5496 (vs. 0.5223 for $|\Gamma|$ alone, 0.500 for majority class); several features are statistically detectable but fall below the pre-declared weakness threshold. On GPT-2 ($n = 22,477$, $\sim 10\times$ the statistical power), the effect does not strengthen: four of eight features remain near-zero despite the added power, and zero of eight reach the within-stratum effect-size threshold. **Verdict: static directional geometry fails to provide a replicated, useful predictor** of the winner; the larger sample provides positive evidence against the hypothesis, not merely an absence of significance.

8.2 DPH: downstream propagation

Reported in Section 7. **Verdict: propagation through depth demonstrably produces substantial winner reversal (roughly one third, cluster-aware), but does not follow a universal, cleanly characterized rule** (e.g. emergence depth varies widely across cases; see stored per-case trajectories).

8.3 FLH: functional/logit-space representation

We repeated the same trace, projecting hidden states to vocabulary space at each layer (logit lens) and measuring KL divergence in output-distribution space rather than residual-stream Euclidean distance, on the identical 250 cases (index-verified 1:1 reuse of the DPH sample, not an independent draw). Cluster-aware paired differences (same resampled cells applied to both analyses within each bootstrap replicate):

$$\Delta_{\text{flip}} = \text{FLH} - \text{DPH} = -0.156, \quad 95\% \text{ CI} = [-0.231, -0.082],$$

$$\Delta_{\text{unresolved}} = \text{FLH} - \text{DPH} = +0.392, \quad 95\% \text{ CI} = [0.294, 0.485].$$

Both intervals exclude zero. **We interpret this as functional space trading fewer observed flips for substantially greater indeterminacy, not as functional space resolving the ambiguity.** Consistent with this reading, among the 56 cases where residual space never stabilizes on the final winner, functional space resolves only 35.7% (95% CI [22.8%, 50.0%]) of them — still a minority. **Verdict: logit-space representation does not resolve the residual-space ambiguity.**

8.4 SDPH: state-dependent propagation

Finally, we tested whether the *same* downstream block responds differently to an *identical* perturbation depending on which policy’s trajectory produced its input state — a targeted, causal swap-intervention audit ($n = 24$ hand-stratified cases, finite-difference scale sweep, validation error 0.0 against natural-forward reproduction). This effect is real: it is present, close to linear in perturbation scale (relative nonlinearity $\sim 4\text{--}9 \times 10^{-4}$ across scales tested), and was established causally, by intervention, not merely correlationally.

This finding must not be read as an explanation for the DPH/FLH reversals, and we state the disconfirming evidence explicitly: state-dependence does *not* track flip status in the direction a mechanistic story would predict. The pooled median state-dependence ratio (flip cases vs. stable cases) is 0.2856 — *flip cases show less* measured state-dependence than stable cases, the opposite of what a “state-dependence causes flips” account would require. Architecture-split, the pattern is inconsistent rather than confirmatory: Pythia shows flip $\approx$ stable (0.0712 vs. 0.0702 median S), while GPT-2 shows flip $\ll$ stable (0.0033 vs. 0.0248). **Verdict: state-dependence is a confirmed, causally demonstrated phenomenon, but a falsified candidate explanation for winner reversal** — it exists, but not in the place or pattern that would explain Sections 7’s result.

8.5 Summary

Figure 4 summarizes the sequence.

Falsification sequence: DVG → DPH → FLH → SDPH

Stage	Hypothesis	Test	Result	Interpretation
DVG	Static directional value-vector geometry (8 cosine/norm features) predicts the downstream winner.	Pooled + within-stratum AUC/correlation on Pythia (n=1,962) and GPT-2 (n=22,477, ~10x the power).	Best CV AUC 0.550 (Pythia) vs. 0.522 for [Gamma] alone; 4/8 features near-zero despite added power on GPT-2; 0/8 reach the within-stratum effect-size threshold.	Failed as a useful predictor; the larger sample strengthens the negative result rather than rescuing it.
DPH	The downstream winner emerges only through propagation across depth.	Cluster-aware causal trace of 250 stratified cases (100 Pythia, 150 GPT-2); cell-level bootstrap, B=10,000.	Pooled flip rate 32.4% [26.1,39.2]; architecture-difference CI crosses zero (Pythia 38.0% [26.4,52.2] vs. GPT-2 28.7% [21.6,36.0]).	Reversal is substantial but heterogeneous; no universal depth rule; architectures not reliably distinguishable.
FLH	Logit/KL-space (functional) representation resolves the residual-space ambiguity.	Same 250 cases reprojected via the logit lens; paired cluster bootstrap against the DPH result.	$\Delta_{\text{flip}} = -0.156 [-0.231, -0.082]$; $\Delta_{\text{unresolved}} = +0.392 [0.294, 0.485]$.	Fewer flips but substantially more unresolved cases; functional space does NOT cleanly resolve the ambiguity.
SDPH	The same downstream block responds differently to an identical perturbation depending on the policy-conditioned base state, explaining the reversals.	n=24 causal swap-intervention audit, finite-difference scale sweep; validation error 0.0.	State-dependence is real and near-linear, but flip/stable median-S ratio = 0.286 (opposite the predicted direction); architecture-inconsistent.	State-dependence CONFIRMED to exist, but FALSIFIED as the explanation for flip status.

Figure 4: The falsification sequence, hypothesis/test/result/interpretation for each stage. Numbers reproduce exactly the conclusions stated in Sections 8.1–8.4 above; no new value is introduced here. The SDPH row deliberately states both that state-dependence is confirmed to exist and that it is falsified as an explanation for the DPH/FLH reversals, per the mandatory distinction maintained throughout this section. Sources: `results/directional_value_geometry/` and `results/directional_value_geometry_gpt2/`, `results/cluster_aware_reanalysis/`, `results/state_dependent_propagation_audit/`; full derivation in `manuscript/figures/FIGURE_PROVENANCE.md`.

9 Discussion

On the simplicity of Theorem 2’s proof. The proof of the Γ boundary is two lines of algebra: dividing one positive quantity by another and taking a monotone log transform. We do not dispute that this is mathematically simple, and we do not claim depth of derivation as the contribution. What Theorem 2 does is convert a question that could otherwise only be answered empirically, case by case (“does the local-perturbation ranking agree with the attention-mass ranking here?”), into a closed-form condition that can be checked exactly and at scale – which is what makes the 482,787-pair verification of Section 5 and the four-stage falsification program of Section 8 possible in the first place. The paper’s substance, in other words, is claimed to lie in the empirical and causal program the boundary condition enables, not in the algebra itself.

The overall picture is that exact local information about a KV-cache eviction decision is genuinely exact — Theorem 1 is an identity, not an approximation, and Theorem 2 gives a provably correct boundary for when two natural rankings of that local information disagree — and that this exactness does not transfer to the downstream question practitioners usually care about. None of the four candidate mechanisms we tested reduces the downstream behavior to something simply characterized in terms of quantities available at eviction time. The one positive causal finding among them (state-dependence, SDPH) is real but points *away from* a simple explanation, since its pattern does not track which cases flip.

On the sequential-hypothesis question: DVG, DPH, FLH, and SDPH were run in sequence, each motivated by the previous stage’s outcome, several of them using pre-declared effect-size

thresholds rather than a single significance gate, and FLH/SDPH partially reuse the DPH sample (explicitly disclosed above, not concealed). We treat this as a directed falsification narrative, not four independent preregistered confirmatory trials, and accordingly do not apply a Bonferroni-style correction across them; we do report cluster-aware, rather than naive i.i.d., uncertainty for every quantity where case non-independence is plausible.

10 Limitations

- Two architectures only (Pythia-160M, GPT-2 124M), both small by current standards; no modern instruction-tuned or much larger model is tested.
- The Pythia grid (55,971 pairs, 5 layers $\times$ 4 heads) and the GPT-2 grid (426,816 pairs, full 12×12) differ in density and coverage; the reported prevalence difference is not a controlled comparison.
- The 250-case downstream sample is stratified by design; reported rates describe the stratified sample’s cluster-aware behavior, not raw unweighted prevalence across all measurable inversions.
- Some subgroup/cluster analyses (notably Pythia-only, 34 clusters, 9 singleton) have wide intervals and should not be over-interpreted; architecture differences in particular are not established (CI crosses zero throughout).
- “Measurable” KL differences are defined relative to a pre-declared numerical noise floor; this establishes the effect is not numerical noise, not that it is practically significant for any downstream task.
- We do not evaluate any policy against deployed eviction systems (H₂O, SnapKV, StreamingLLM, etc.) on task quality; this paper makes no claim about which policy one should deploy.
- Logit-lens-based measurements carry float32 reconstruction noise (bounded at 2.2×10^{-5} against true final-layer KL in our validation), handled by a diagnostically justified tolerance (10^{-4}), not a tolerance chosen to fit the result.

11 Conclusion

We derived an exact, closed-form pairwise boundary for when attention-mass and exact local-output-perturbation rankings of KV-cache eviction candidates disagree, built on a singleton identity that we explicitly do not claim as novel (it is algebraically identical to prior KeepKV work [1]). Across two architectures, such disagreements are common, almost always change the eviction decision, and almost always produce a measurable downstream effect — yet the locally preferred choice is overturned by later computation in roughly a third of cluster-aware sampled cases, and a four-stage falsification program finds no simple predictor of when. Exact local correctness, in other words, is not the same thing as downstream predictive power, and this paper’s evidence is a specific, quantified account of that gap for one narrow but concrete setting, not a general claim about transformers.

A Figure Status

This appendix records where each main-paper figure lives and how it was produced.

1. **Exact local eviction geometry and the $\Gamma = 0$ boundary** — Figure 1, Section 4. Schematic (synthetic numeric example run through the real formulas of Theorems 1/2); no stored result file is

used. Region labels compare only which of two candidates has the larger exact local perturbation ($d_i > d_j$ vs. $d_i < d_j$), not which eviction policy wins downstream.

2. **Inversion** $\rightarrow$ **set-change** $\rightarrow$ **measurable-KL chain** — Figure 2, Section 6. Data-derived from `results/full_grid_mechanistic_analysis/` (Pythia) and `results/cross_architecture_full_grid/` (GPT-2), recomputed directly by the generating script rather than transcribed. Reports Section 6 statements (1)–(3) only; statement (4) is Figure 3.
3. **Representative downstream trajectories** — Figure 3, Section 7. Data-derived from `results/downstream_propagation_sample/cases.json`; one case per category selected by a fixed median-magnitude rule, documented in full in `manuscript/figures/FIGURE_PROVENANCE.md`.
4. **Falsification summary (DVG $\rightarrow$ DPH $\rightarrow$ FLH $\rightarrow$ SDPH)** — Figure 4, Section 8. Renders, with no new numbers, the conclusions already stated in Section 8’s four subsections; supersedes an earlier 3-column LaTeX table with a 4-column (hypothesis/test/result/interpretation) layout.

Generation script: `manuscript/figures/generate_figures.py` (requires only already-stored result files and `matplotlib`; performs no model forward pass). Full per-figure provenance — exact source file, result keys, filters, case IDs, and any aggregation — is in `manuscript/figures/FIGURE_PROVENANCE.md`.

B Supplementary Material Not Reproduced in the Main Text

The main text is deliberately restricted to the theorem-to-downstream narrative. The following material exists in the project’s stored results but is intentionally not reproduced in the main text; a future appendix pass should draw directly from these files rather than re-deriving numbers:

- **Full adapter validation detail** beyond the summary figures quoted in Section 5: `results/cross_architecture_generalization/report.txt`, `tests/test_gpt2_adapter.py`.
- **Full architecture grids** (per-cell counts, strongest pairs, percentile tables): Pythia in `results/full_grid_mechanistic_analysis/summary.json`; GPT-2 in `results/cross_architecture_full_grid/summary.json` (report at `results/cross_architecture_full_grid/report.txt`).
- **Per-stratum heterogeneity** (Gini coefficient of inversions across cells, strongly mass-/local-favoring strata, concentration-vs-prevalence correlations): `results/cross_architecture_full_grid/report.txt`.
- **The 36-cell GPT-2 pilot grid**, used only for adapter validation in the main text (Section 5): full pilot report at `results/cross_architecture_generalization/report.txt`. Its own prevalence estimate (6.62%, a third, smaller grid with different layer/head coverage than either headline grid) is recorded there but is not a headline result of this paper.
- **Extended DVG feature detail** (all 8 directional features, pooled/within-stratum statistics, tail-case audits, minimal multivariate models): `results/directional_value_geometry/report.txt`, `results/directional_value_geometry_gpt2/report.txt`.
- **FLH residual-vs-functional cross-tabulation** (the full 2 $\times$ 2 breakdown underlying the “resolves 35.7% of never-consistent cases” statistic) and per-layer KL/JS/L2/cosine detail: `results/functional_propagation_audit/report.txt`, `results/functional_propagation_audit/cases.json`.
- **SDPH linearity controls** (state-dependence S at multiple perturbation scales, attention-vs-MLP component audit, per-architecture breakdowns): `results/state_dependent_propagation_audit/report.txt`.
- **Cluster-bootstrap sensitivity analyses** (alternate cluster definitions: layer $\times$ head collapsing architecture and prompt; architecture $\times$ prompt collapsing layer and head): the `sensitivity_alternate_cluster_definitions` key of `results/cluster_aware_reanalysis/summary.json`.

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ematical and empirical results, and retained responsibility for all scientific claims and conclusions. Claude (Anthropic) was used extensively for repository-based implementation, computational experimentation, formal auditing, statistical analysis, literature-assisted review, figure generation, and manuscript development. ChatGPT (OpenAI) was used for research planning, mathematical and conceptual analysis, interpretation of experimental results, critical evaluation of claims, manuscript strategy, and iterative synthesis across the research program. Both AI systems contributed substantially to the research process as interactive computational and reasoning tools. The author reviewed and approved the resulting mathematics, experiments, code, interpretations, and manuscript and assumes responsibility for the final work.

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